

Equivalent Latitude

Concept, calculation, and comparison

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What is Equivalent Latitude?

Features and importance

- Equivalent Latitude ϕ_e is a quasi-Lagrangian coordinate.
- It is defined as the latitude that encloses a circle centered at the pole with the same area as the isoline of an atmospheric field (e.g. PV).

$$\phi_e = \arcsin\left(\frac{1 - A}{2\pi R^2}\right)$$

Moves with air masses – adiabatic transport is eliminated -> e.g. breaking of Rossby-Waves

Separates dynamic variability and chemical changes -> e.g. photochemical ozone depletion in the vortex

Used for tracer climatology and comparison with model data -> e.g. when observations are sparse

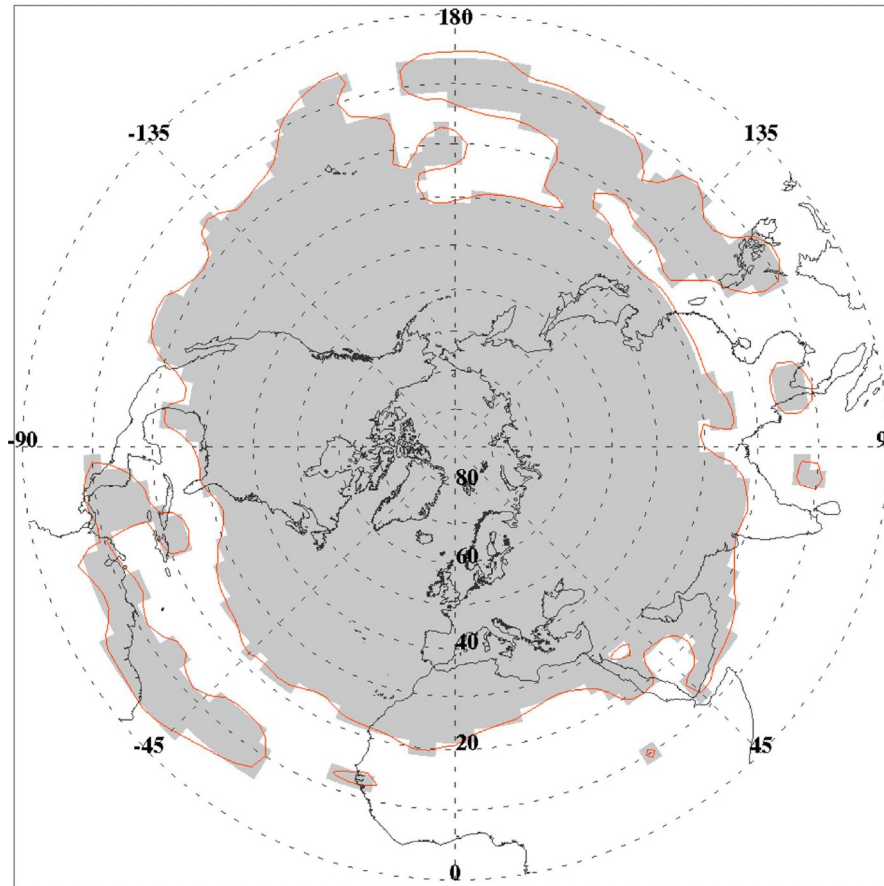
Piecewise-Constant vs. Region of Interest method

PW-Constant

- PV is assumed to be constant within each grid cell.
- All grid cells whose PV value is below a threshold value are identified.
- The areas of all selected cells are added together, and the equivalent latitude is calculated from the total area.

ROI

- PV is not considered constant within each grid cell – variation within the cell is allowed.
- Contour mapping function - Plotting/interpolating PV isolines.
- Geometric areas are calculated and converted to equivalent widths.



2 PVU on an isentropic surface of 380 K.
(Anel et al. 2013)

Motivation

- Hardly any Python packages available for EqLat calculation (only pyvortex, NASA JETPAC for IDL)
- Goal 1: Implementation and comparison of Piecewise-Constant and Region of Interest methods in python -> community package.
- Goal 2: Comparison of MERRA-2 and ERA5 reanalysis in terms of EqLat.
- Goal 3: First look at a potential new approach (SWOOSH product).
- Goal 4: Machine Learning approach to derive EqLat for Radiosondes.

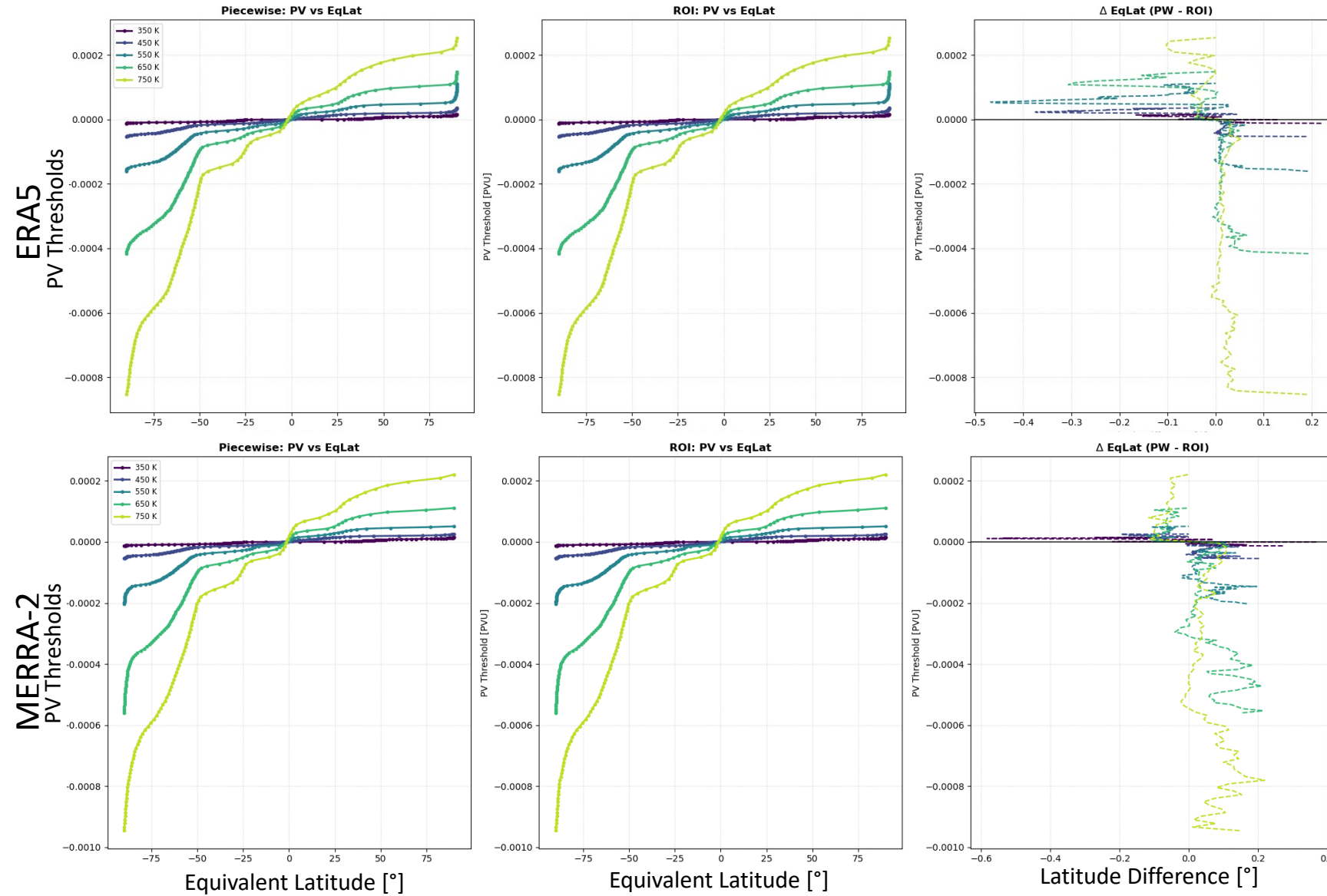
The Reanalyse Products

MERRA-2 (NASA)	ERA5 (ECMWF)
1980 - today	1940 - today
0.5° x 0.635° (lat x lon)	0.25° x 0.25° (lat x lon)
3 – hourly	1 - hourly

- Both are publicly available on pressure fields – must be interpolated on isentropic fields.
- ERA5 via CDS API (account needed).
- MERRA-2 via the Earthdata- system (account needed).
- For this analysis: MERRA-2 and ERA5 data for the year 2020, but only for 12 UTC (still a lot of data). For the selected days, full-time coverage.

The intermediate step – looking at PV and Eqlat on isentropic levels

2020-06-30 12:00 UTC



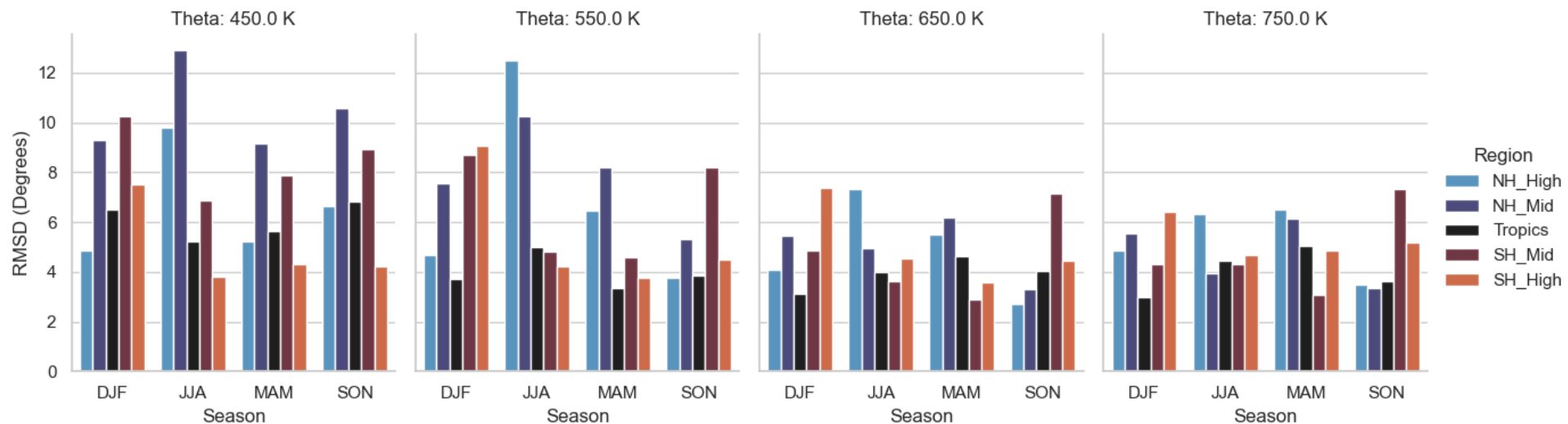
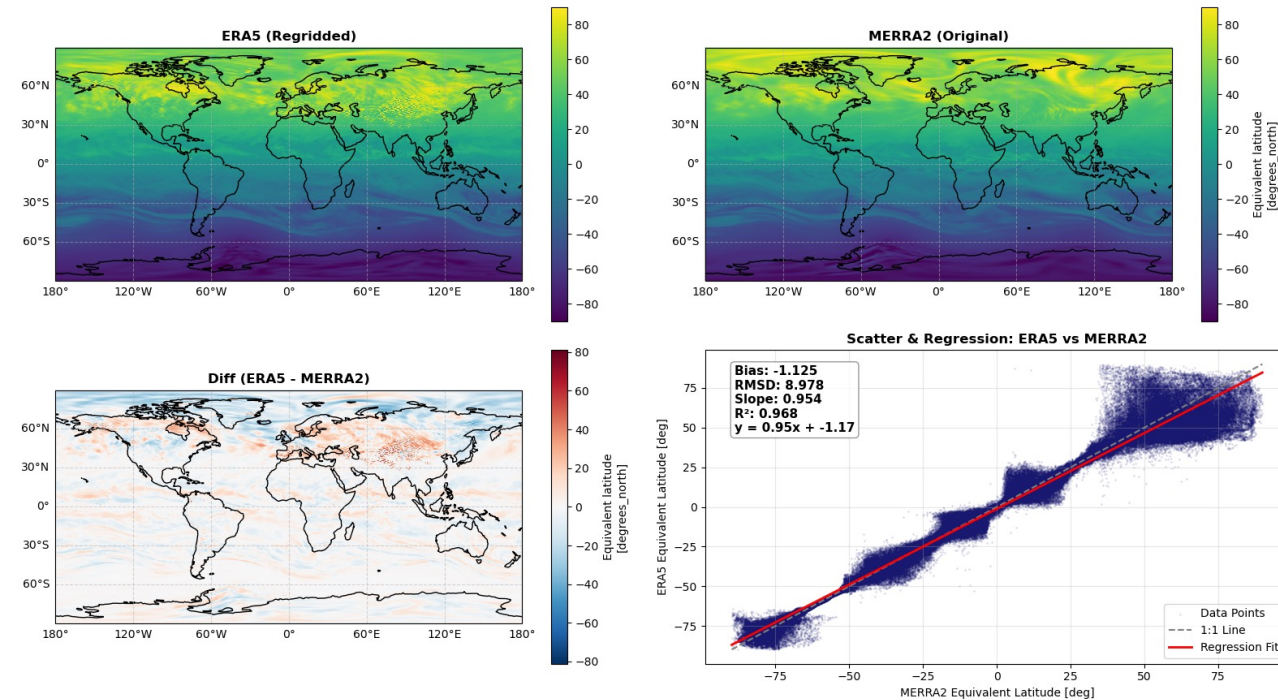
- Largest differences in the summer hemisphere where gradients are weak.
- Differences are also greater in the polar regions—small-scale features.
- Differences are smaller in ERA5 than in MERRA-2, possibly due to grid size.

TH: 350		
ERA5	PW: 69 ms	ROI: 514 ms
TH: 450		
ERA5	PW: 65 ms	ROI: 436 ms
TH: 550		
ERA5	PW: 63 ms	ROI: 367 ms
TH: 650		
ERA5	PW: 62 ms	ROI: 344 ms
TH: 750		
ERA5	PW: 62 ms	ROI: 311 ms

ERA5 vs. MERRA-2. (PW)

- Usually larger differences in summer hemisphere mid- and high latitudes.
- Winter hemisphere typically shows two-part variability, occurring in high latitudes and lower mid-latitudes.
- Consistent larger difference in SH spring – Vortex break up and filamentation.

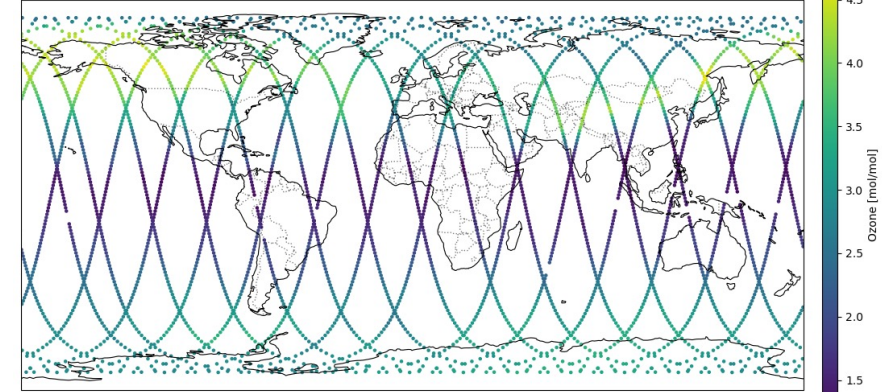
Comparison at 550K | 2020-06-30 12:00



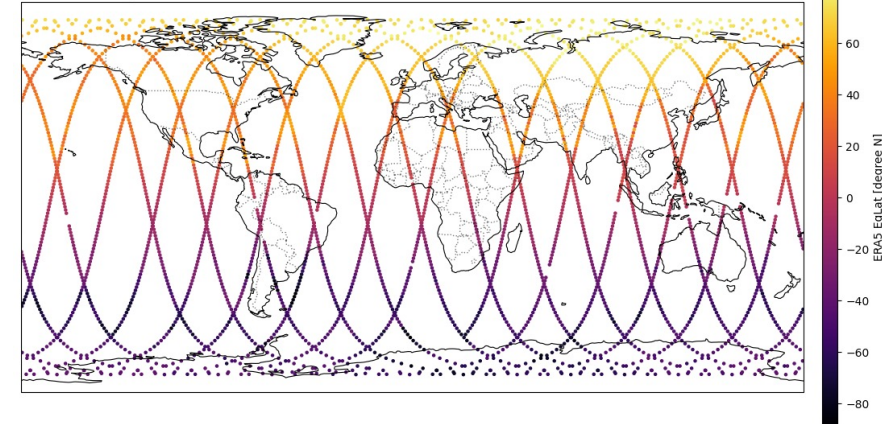
ERA5 and MERRA-2 Compactness Test

- Evaluate the latitudinal consistency of the ozone data.
- MLS stellite data used with quality filtering.
- Deriving zonal mean of MLS O_3 in latitude and equivalent latitude.

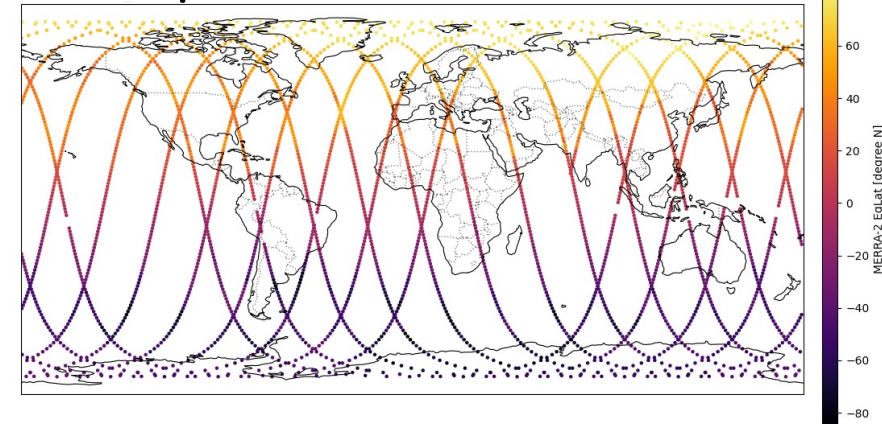
MLS O3 at 50 hPa 2020-01-01



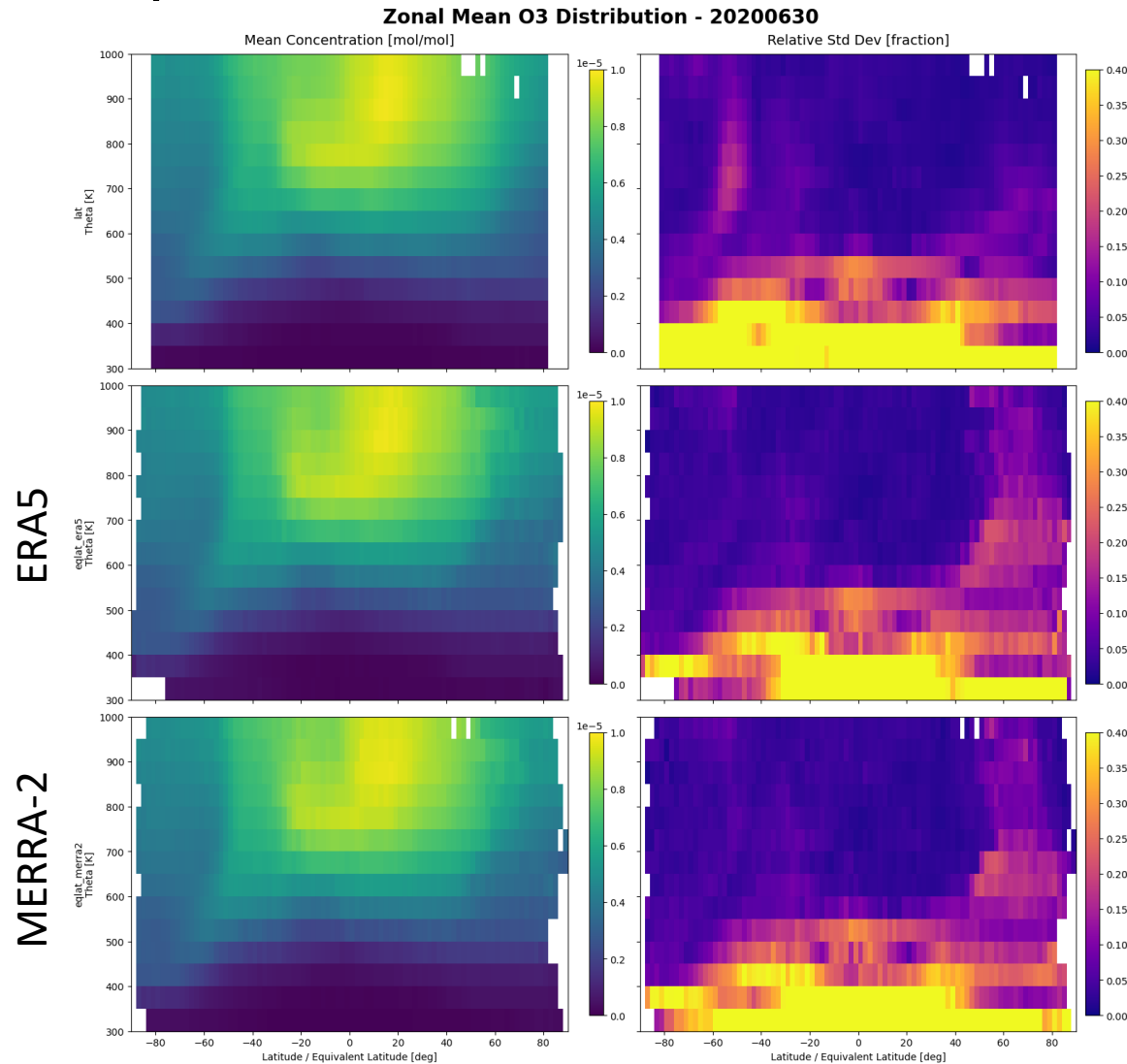
ERA5 eqlat at 50 hPa



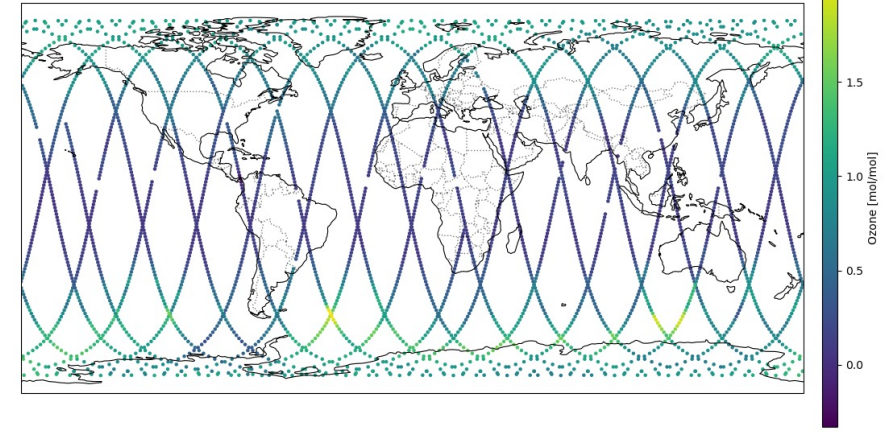
MERRA-2 eqlat at 50 hPa



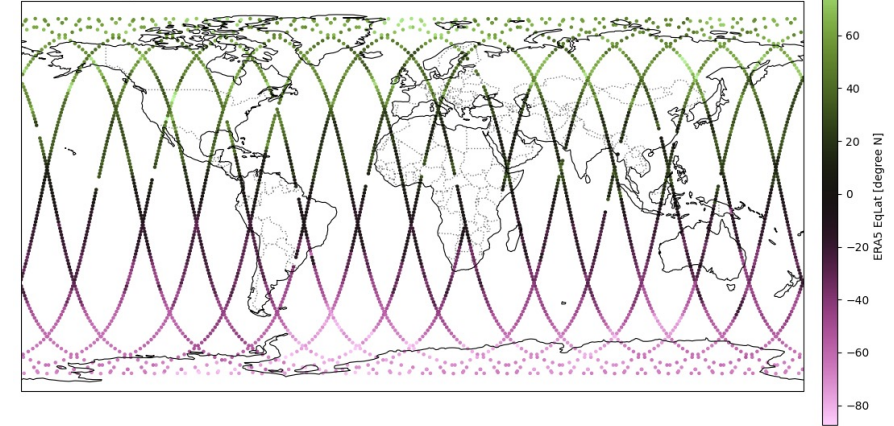
ERA5 and MERRA-2 Compactness Test



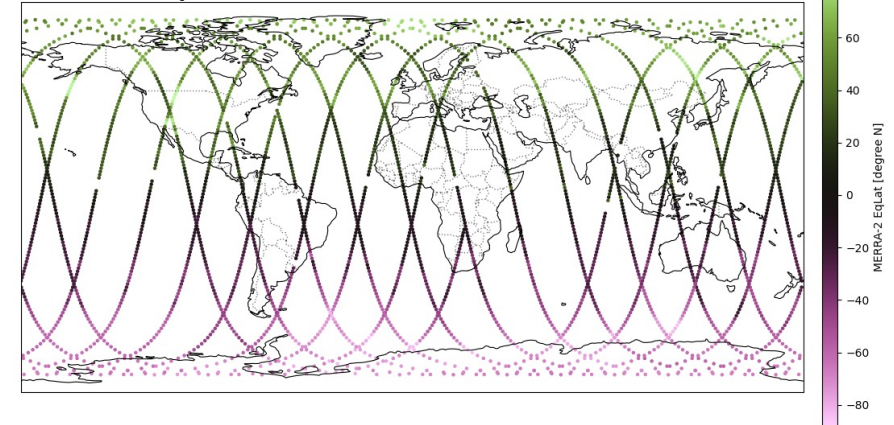
MLS O3 at 100 hPa



ERA5 eqlat at 100 hPa

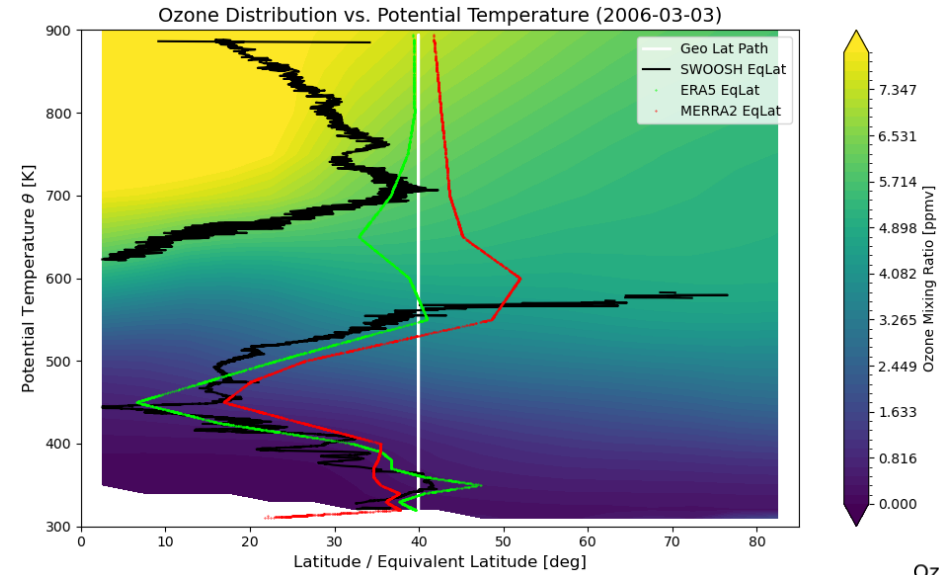


MERRA-2 eqlat at 100 hPa

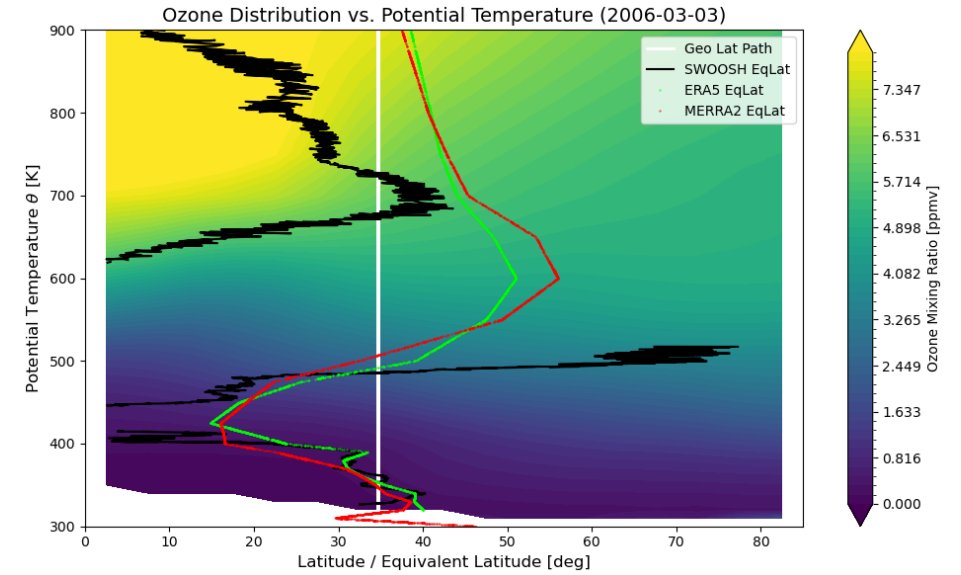


SWOOSH- ERA5 – MERRA-2 same day, different location

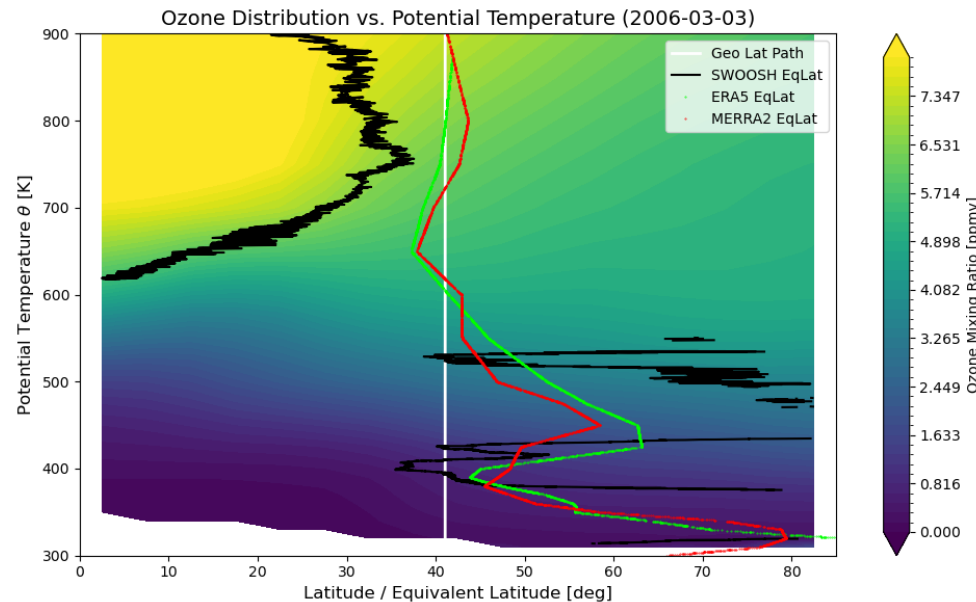
Boulder



Huntsville



Trinidadhead



Summary

- PW and ROI method show only small differences (0.1° to around 0.6°), but PW method is faster (at least in python at the moment).
- The greatest differences are found in regions with weak PV gradient and/or near the poles, where the areas become smaller.
- MERRA-2 and ERA5 overall comparable, but show larger differences in respective summer hemispheres (weak PV gradients) and lower isentropic levels (more structure).
- PV-based Equivalent Latitude seems to perform better in winter hemisphere, where PV gradient on isentropes are stronger.
- SWOOSH monthly zonal mean ozone distribution - not really satisfying. Next step is look at N_2O from SWOOSH (waiting for data) and move to Machine Learning approach.

